

Geometry of Recursive Knots:

Iterated Cables by Frame Offsets, Their Thickness and Ropelength

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Abstract

We study a concrete family of *recursive knots*: starting from a round circle K_0 , K_d is the (p_d, q_d) torus pattern on the tube of radius r_d around K_{d-1} , placed with a rotation-minimising (Bishop) frame closed by a linear twist; the question is how much rope (length divided by thickness) such nested knots need. We prove that K_d is embedded and smooth when $r_d < \text{thick}(K_{d-1})$; that the closed frame has linking number $\text{Wr } K_{d-1} - \alpha_d/2\pi$ with its core, so K_d is an explicit cable of K_{d-1} ; two-sided length bounds; and, for $p_d = 2$, $\text{thick}(K_d) \leq r_d$, so $\text{Rop}(K_d) \geq L(K_d)/r_d$. For the lower bound $\text{thick}(K_d) \geq \min(\text{thick } K_{d-1} - r_d, c r_d \sin(\pi/p_d))$ with $p_d = 2$, $r_d \leq \text{thick}(K_{d-1})/2$: in the default family (2, 3) every doubly critical pair of K_d with distinct base points is at distance $\geq \min(2(\text{thick } K_{d-1} - r_d), r_d)$ at every depth, so the bound with $c = 1/2$ reduces to $\min\text{Rad}(K_d) \geq r_d/2$, which we prove at depth one by hand and at depth two by a computer-assisted argument (interval arithmetic on the explicit base K_1 : $\min\text{Rad}(K_2) \geq 0.842 r_2$ for every $f \leq 1/2$ and every choice of phases). For $d \geq 3$ it follows from a bound (H_3) on derivatives of the speed and of the curvature vector of K_{d-1} , and fails for an arbitrary base without one. We show that the speed and curvature of K_d are controlled by the (H_3) data of K_{d-1} and that for $d \geq 3$ a growth condition on these data suffices, but that $\mu(K_d)$, the size of the derivative of the curvature vector, cannot be bounded by any locally bounded function of the (H_3) data of an arbitrary base (one derivative is lost per depth in this sense; in the family μ grows slowly on the computed levels). (H_3) uniformly in d , hence the bound for $d \geq 3$, remains open, although every condition holds on the polygons (grid evaluation, uncertified). Under the thickness bound (proved for $d \leq 2$), $\text{Rop}_d \leq 2\pi\Lambda_{\text{Rop}}^d$ with an explicit Λ_{Rop} . Explicit polygons with up to 8192 vertices give $L/\tau = 38.4, 155.6, 622.7$ at depths 1, 2, 3 for (2, 3) and $f = 1/2$, stable to 0.03% under doubling of the resolution: numerically verified upper bounds for explicit polygons, not certified smooth thicknesses. No stationarity, optimality or recursive flow law is claimed.

1 Introduction and status

The research line. Line RTK001 (“Geometry of recursive knots”) asks how nested knots—knots tied on the tube of other knots, recursively—can be represented as tubes without self-intersection, and how much length that costs. Its public record states one finding (a feasible upper bound at depth one) and an explicit scope restriction: neither stationarity nor optimality of any configuration is certified, and no general recursive flow law has been established. No manuscript of the line existed before this draft; the family below was fixed from the record. Four internal reviews have been applied (v0.2–v0.5, 30/09–03/10/2026): the thickness hypothesis of v0.1 was false for an arbitrary base (Remark 3.13); the same-strand case is proved (Proposition 3.10); the frame normalisation is fixed in Definition 2.1; a partial result on the uniformity of (H_3) is added (Lemma 3.14–Proposition 3.17); and v0.5 adds a computer-assisted proof of the case $d = 2$ (Proposition 3.19), a joint scan of the two frame normalisations that affect K_3 , and states Conjecture 3.25 uniformly in the phases.

What is done here. We (i) fix a definite family of recursive knots (Section 2); (ii) prove embeddedness, the framing used, two-sided length bounds, an upper bound on the thickness and most of the lower bound, isolating what is not proved in one hypothesis on the family (Section 3); (iii) compute explicit polygonal configurations at depths 1 to 3, their polygonal thickness and ropelength, with a resolution check (Section 4); (iv) tabulate every claim with its status (Section 5). Lower bounds for general knots are only quoted from the literature (Remark 3.24); the only lower bound we derive is internal to the family (Proposition 3.5).

2 Setting

2.1 Thickness and ropelength

Let $K \subset \mathbb{R}^3$ be a closed $C^{1,1}$ curve of length $L(K)$. Its *thickness* $\text{thick}(K)$ is the supremum of ρ such that the open tube of radius ρ about K is embedded (normal injectivity radius). Two equivalent descriptions are classical. Litherland, Simon, Durumeric and Rawdon [11] show, for C^2 curves (extended to the $C^{1,1}$ case in [3]; all curves K_d below are C^∞),

$$\text{thick}(K) = \min(\min\text{Rad}(K), \tfrac{1}{2} \text{dcsd}(K)), \quad (1)$$

where $\min\text{Rad}$ is the minimal radius of curvature and dcsd the *doubly critical self distance*, the minimal distance between pairs $x \neq y$ of K whose chord is perpendicular to K at both ends. Gonzalez and Maddocks [7] show $\text{thick}(K) = \inf_{x \neq y} \rho_{pt}(x, y)$, with $\rho_{pt}(x, y) = |x - y| / (2 \sin \angle(T_x, y - x))$ the radius of the circle through y tangent to K at x . The *ropelength* is $\text{Rop}(K) = L(K) / \text{thick}(K)$; it is scale invariant, $\text{Rop}(K) \geq 2\pi$ for every closed curve (Fenchel's theorem [5], total curvature $\geq 2\pi$, together with $\kappa \leq 1/\text{thick}$), with equality only for the circle, and $\text{thick}(K) \leq \min\text{Rad}(K)$ always.

2.2 Frames, holonomy and the closing twist

Let K be a closed C^2 regular curve parametrised on $[0, 2\pi)$ (not necessarily by arclength), with unit tangent T and curvature κ . A *rotation-minimising* or Bishop frame [1] is an orthonormal frame (T, N_0, B_0) with $N'_0 = -\kappa_1 |K'| T$, $B'_0 = -\kappa_2 |K'| T$, $\kappa_1^2 + \kappa_2^2 = \kappa^2$; it is unique up to a constant rotation in the normal plane (fixed in Definition 2.1) and is in general *not* periodic: after one turn the transported normal is rotated by a holonomy angle $\alpha \in (-\pi, \pi]$. We close it by the linear twist $\omega(t) = -\alpha t / 2\pi$, i.e. $N = \cos \omega N_0 + \sin \omega B_0$, $B = T \times N$. The closed frame satisfies

$$N' = -\tilde{\kappa}_1 |K'| T + \omega' B, \quad B' = -\tilde{\kappa}_2 |K'| T - \omega' N, \quad |\omega'| = \frac{|\alpha|}{2\pi} \leq \frac{1}{2}, \quad (2)$$

with the rotated curvature components $\tilde{\kappa}_1 = \kappa_1 \cos \omega + \kappa_2 \sin \omega$, $\tilde{\kappa}_2 = -\kappa_1 \sin \omega + \kappa_2 \cos \omega$ (same norm κ); below we write κ_1, κ_2 for whichever frame is in use.

2.3 The family of recursive knots

Definition 2.1 (iterated cable by frame offsets). Let K_0 be the circle of radius $R_0 = 1$. For $d \geq 1$ choose coprime integers $p_d \geq 2$, $q_d \geq 1$, a radius $r_d > 0$ and a phase φ_d . Reparametrise K_{d-1} to period 2π by the linear change $t \mapsto t/p_{d-1}$ (so that K_{d-1} is traversed at parameter speed $|K'_{d-1}|$, not at unit speed), let (N_{d-1}, B_{d-1}) be its closed Bishop frame, normalised by $N_{d-1}(0) =$ the unit projection of e_z onto the normal plane at $K_{d-1}(0)$ (of e_x if $|T(0) \cdot e_z| > 0.9$), and set for $t \in [0, 2\pi p_d)$

$$K_d(t) = K_{d-1}(t \bmod 2\pi) + r_d [\cos \theta_d(t) N_{d-1}(t) + \sin \theta_d(t) B_{d-1}(t)], \quad \theta_d(t) = \frac{q_d}{p_d} t + \varphi_d. \quad (3)$$

Rotating $N_{d-1}(0)$ by an angle ϑ amounts to replacing φ_d by $\varphi_d + \vartheta$, so a choice of normalisation is a choice of phases; for $p_d = 2$, q_d odd, $\varphi_d + \pi$ gives the same curve. The default is $(p_d, q_d) = (2, 3)$, $\varphi_d = 0$ at every depth; r_d is a fraction f of the thickness of K_{d-1} (the guaranteed lower bound ρ_{d-1} in the theory, the measured polygonal value in the experiments).

K_1 is the (p_1, q_1) torus knot on the round torus of radii $(1, r_1)$; the knot type of K_d for $d \geq 2$ depends on the framing, which Lemma 3.2 makes explicit. All results of Section 3 hold for every choice of $N_{d-1}(0)$ and φ_d ; lengths and thicknesses of the computed levels do not depend on it at the printed precision, but the margins of Conjecture 3.25 do (Section 4). The pattern angle advances in the parameter t , not in arclength; since the parametrisation of K_{d-1} is the one inherited from the construction, the *pitch* of the strands relative to the base (the quantity h of Section 3) is a property of the family, not a free parameter.

2.4 Polygonal measurements

Numerically a curve is a closed polygon $P = (x_0, \dots, x_{N-1})$. We use $L(P) = \sum |x_{i+1} - x_i|$; $\min\text{Rad}(P)$, the minimum circumradius of consecutive triples; $\text{dcsd}(P)$, the minimal distance $|x_i - x_j|$ over *doubly*

critical vertex pairs, i.e. pairs where $|x_i - x_j|$ is a local minimum in i and in j (the diagonal $D_{ii} = 0$ is kept in the neighbourhood test, which excludes near-diagonal pairs); and the proxy

$$\tau(P) = \min(\min\text{Rad}(P), \frac{1}{2} \text{dcsd}(P)), \quad \text{Rop}(P) = L(P)/\tau(P). \quad (4)$$

The local-minimum test could miss doubly critical pairs of saddle type; as a safeguard we also compute $\tau_{pt}(P) = \min_{i \neq j} |x_j - x_i|^2 / (2|T_i \times (x_j - x_i)|)$ with central-difference tangents T_i , which uses no criticality test. Neither τ nor τ_{pt} is a certified thickness of a smooth curve (for the thickness of polygons themselves see [13]); they are the quantities of (1) evaluated on an explicit polygon, and their stability under doubling of N is the main evidence offered that they approximate the smooth values. Vertex pairs may overestimate the polygon–polygon distance; the minimal segment–segment distance over pairs of edges with cyclic index separation $> N/10$, divided by the vertex dcsd, is 0.99994, 1.00000 ($d = 1, f = 0.25, 0.5$) and 0.99993 ($d = 2, f = 0.5$), and near the critical pair the ratio is ≥ 0.99998 in every level examined (`experiments/aux_checks.py`). Writhe is the Gauss double sum over pairs of edges (midpoint rule); its accuracy is discussed in Section 4.

3 Results

Throughout, $K = K_{d-1}$ is a closed C^∞ regular curve of thickness $\tau = \text{thick}(K) > 0$, hence of maximal curvature $\kappa_{\max} \leq 1/\tau$, carrying the parametrisation of period 2π inherited from Definition 2.1, with speed $v = |K'| \in [v_{\min}, v_{\max}]$; K_d is built by (3) with $r = r_d, p = p_d, q = q_d$, and $f = r/\tau$.

Lemma 3.1 (embedding). *[proved here] If $r < \tau$ then K_d is a regular C^∞ closed curve, injective on $[0, 2\pi p)$, contained in the torus $\partial\{x : \text{dist}(x, K) < r\}$.*

Proof. Smoothness: the Bishop frame of a C^∞ curve is C^∞ (it solves a linear ODE with smooth coefficients), and (3) is a smooth expression in it; periodicity of K_d in t with period $2\pi p$ holds because $\theta_d(t + 2\pi p) = \theta_d(t) + 2\pi q$ and the closed frame has period 2π . Regularity: differentiating (3) with (2),

$$K'_d = |K'| (1 - r(\kappa_1 \cos \theta + \kappa_2 \sin \theta)) T + r(\theta' + \omega')(-\sin \theta N + \cos \theta B), \quad (5)$$

and $|\kappa_1 \cos \theta + \kappa_2 \sin \theta| \leq \kappa \leq 1/\tau$, so the tangential coefficient is at least $|K'|(1 - r/\tau) > 0$. Injectivity: let $t_1 \neq t_2$ in $[0, 2\pi p)$. If $t_1 \not\equiv t_2 \pmod{2\pi}$, the two points lie in the normal discs of radius $r < \tau$ at distinct points of K ; by definition of thickness these discs are disjoint, so the points differ. If $t_1 \equiv t_2 \pmod{2\pi}$, then $t_1 - t_2 = 2\pi k$ with $0 < |k| < p$ and $\theta_d(t_1) - \theta_d(t_2) = 2\pi qk/p \notin 2\pi\mathbb{Z}$ because $\gcd(p, q) = 1$; the two points lie in the same normal disc at the same radius $r > 0$ but at different angles, hence differ. Containment in the torus is immediate from (3). $\square$

The constant $1/2$ in $r \leq \tau/2$, used from Lemma 3.3 on, is what the estimates below need; embeddedness only needs $r < \tau$.

Lemma 3.2 (the framing used, and the knot type). *[proved here, from classical results] Let $\alpha \in (-\pi, \pi]$ be the holonomy of the Bishop frame of K . Then the closed frame (N, B) is a framing of K with linking number $n := \text{Lk} = \text{Wr } K - \alpha/2\pi \in \mathbb{Z}$; in particular $\alpha \equiv 2\pi \text{Wr } K \pmod{2\pi}$, and $n = \text{round}(\text{Wr } K)$ whenever $\text{Wr } K \notin \mathbb{Z} + \frac{1}{2}$ (if $\text{Wr } K \in \mathbb{Z} + \frac{1}{2}$ then $\alpha = \pi$ and $n = \text{Wr } K - \frac{1}{2}$). With the convention that a (p, q') -cable of K winds p times along K and q' times around it relative to a framing of linking number 0 (the Seifert framing), K_d is the $(p, q + pn)$ -cable of K [10]. In particular K_1 is the torus knot $T(p_1, q_1)$.*

Proof. By the Călugăreanu–White–Fuller formula [4, 6, 14], $\text{Lk} = \text{Tw} + \text{Wr}$ for any framing. The Bishop frame has zero twist density; the closing rotation ω adds total twist $\text{Tw} = \frac{1}{2\pi} \int \omega' dt = -\alpha/2\pi \in [-\frac{1}{2}, \frac{1}{2})$. Hence $\text{Lk} = \text{Wr} - \alpha/2\pi$, and since $\text{Lk} \in \mathbb{Z}$ the congruence and the rounding rule follow. The pattern (3) winds q times around the meridian relative to the frame (N, B) while going p times along K ; relative to a framing of linking number 0 it winds $q + pn$ times, which is the stated cable. For K_0 a planar circle, $\text{Wr} = 0 = \alpha$ and K_1 is the standard torus knot. $\square$

Lemma 3.3 (length, two-sided). *[proved here] If $r \leq \tau/2$ then*

$$p L(K) (1 - r\kappa_{\max}) \leq L(K_d) \leq p L(K) (1 + r\kappa_{\max}) + r(2\pi q + p|\alpha|) \leq p L(K) (1 + r/\tau) + r(2\pi q + \pi p). \quad (6)$$

Proof. From (5) and the triangle inequality, $|K'_d| \leq |K'| (1 + r\kappa) + r(|\theta'| + |\omega'|) = |K'| (1 + r\kappa) + r(q/p + |\alpha|/2\pi)$. Integrate over $[0, 2\pi p)$: the first term gives $p \int_0^{2\pi} |K'| (1 + r\kappa) dt \leq p L(K) (1 + r\kappa_{\max})$, the second $r(2\pi q + p|\alpha|)$. Then use $\kappa_{\max} \leq 1/\tau$ and $|\alpha| \leq \pi$. For the lower bound, $|K'_d|$ is at least its tangential component $|K'| (1 - r\kappa) \geq |K'| (1 - r\kappa_{\max})$; integrate. $\square$

Remark 3.4. The term $p|\alpha|r \leq \pi pr$ from the closing twist, absent from a naive count “ p copies plus q meridians”, is needed in the proof because the Bishop frame of a non-planar K does not close; numerically the bound without it also holds in all 36 levels (minimal ratio bound/ L 1.255).

Proposition 3.5 (upper bound on the thickness and a lower bound on the ropelength, $p = 2$). *[proved here] Let $p = 2$ (q odd) and $r < \tau$. Then $\text{thick}(K_d) \leq r$, hence $\text{Rop}(K_d) \geq L(K_d)/r \geq pL(K)(1 - r\kappa_{\max})/r$. In the family with $r_d = f \text{thick}(K_{d-1})$ and $f \leq 1/2$ this gives $\text{Rop}(K_d) \geq \frac{p(1-f)}{f} \text{Rop}(K_{d-1})$, so $\text{Rop}(K_d) \geq 2\pi(p(1-f)/f)^d$.*

Proof. The two points $K_d(t)$ and $K_d(t + 2\pi)$ lie in the same normal disc of K at antipodal angles (θ and $\theta + \pi q$, q odd), so their chord is $-2r(\cos \theta N + \sin \theta B)$, orthogonal to T and to $-\sin \theta N + \cos \theta B$, hence by (5) orthogonal to K'_d at both ends: the pair is doubly critical at distance $2r$, so $\text{dcsd}(K_d) \leq 2r$ and $\text{thick}(K_d) \leq r$ by (1) (equivalently, $\rho_{pt} = r$ for this pair). The rest is Lemma 3.3 with $\kappa_{\max} \leq 1/\tau$, $r = f\tau$ and $\text{Rop}(K_{d-1}) = L(K_{d-1})/\tau$. $\square$

For the default family at $f = 1/2$ the internal lower bound is $2\pi \cdot 2^d$, i.e. 12.6, 25.1, 50.3 at $d = 1, 2, 3$, against the measured 38.4, 155.6, 622.7; with the measured $L(K_d)$ the bound $L(K_d)/r_d$ is attained whenever $\text{thick}(K_d) = r_d$ (Section 4). The lower bound of (6) holds in all 36 measured levels (36 of 36).

Thickness of K_d : what is proved and what is not. Write $s_p = 2r \sin(\pi/p)$ for the chord between adjacent strands in a normal disc of K (the p points $K_d(t + 2\pi k)$, $k = 0, \dots, p-1$, sit at the p equally spaced angles $\theta + 2\pi qk/p$, since $\gcd(p, q) = 1$).

Proposition 3.6 (two of the three cases). *[proved here] If $r \leq \tau/2$: (a) two points of K_d in the same normal disc of K are at distance $\geq s_p$; (b) two points of K_d whose base points on K are at distance $\geq 2\tau$ are at distance $\geq 2(\tau - r)$.*

Proof. (a) They are at the same radius r and at angles differing by $2\pi qk/p$, $0 < k < p$, so their distance is $2r |\sin(\pi qk/p)| \geq 2r \sin(\pi/p) = s_p$ because $qk \not\equiv 0 \pmod{p}$. (b) Triangle inequality: $|K_d(t_1) - K_d(t_2)| \geq |K(t_1) - K(t_2)| - 2r$. $\square$

Not covered by Proposition 3.6 are the pairs with $0 < |K(t_1) - K(t_2)| < 2\tau$, and the curvature of K_d . These include nearby base points (the helix geometry of p strands on a bent tube) and base points on different arcs of K , e.g. across the hole of a torus: at $f = 1/2$, $d = 1$ the doubly critical vertex pair that realises dcsd has base points 103.7° apart on K_0 , at distance $1.573 < 2\tau_0 = 2$ (`aux_checks.py`), while for $f \leq 0.35$ it is the antipodal pair of one normal disc. For a straight tube the p strands are helices with pitch parameter $h = |K'|p/q$ and the antipodal pair ($p = 2$) is a local minimum of the strand distance exactly when $h \geq r$; at depth one the relevant pitch on the inner equator is $(1-r)p/q$, and $(1-r)p/q \geq r$ iff $r \leq p/(p+q) = 0.4$ for (2, 3), consistent with $\text{thick}(K_1) = r_1$ for $f \leq 0.35$ and $\text{thick}(K_1) < r_1$ for $f = 0.5$.

Notation for the local case. Let (T, N_0, B_0) be the (non-periodic) Bishop frame of K and α its holonomy, and set

$$m := \frac{q}{p} - \frac{\alpha}{2\pi}, \quad h_{\min} := \frac{v_{\min}}{m}, \quad \Theta_{\min} := \frac{h_{\min}}{\tau}, \quad \zeta_{\min} := \frac{(1-f)h_{\min}}{r}, \quad f = \frac{r}{\tau} \leq \frac{1}{2}.$$

In the parallel frame the strand of (3) turns at the constant rate m (its angle is $\psi = \theta + \omega = mt + \varphi$), so $h = v/m$ is the pitch parameter above ($|K'|p/q$ when $\alpha = 0$) and $\Theta_{\min}$ is the smallest pitch in units of the base thickness. For two points $X_i = K(t_i) + rU_i$ of K_d with distinct base points, let σ be the length of the *shorter* base arc between $K(t_1)$ and $K(t_2)$, $\delta = \sigma/\tau$, s the base-parameter difference along that arc and $u = m|s|$ the relative turning of the strands along it. The integer $k = (t_2 - t_1 - s)/2\pi \pmod{p}$ is the strand index of X_2 relative to X_1 *along that arc* (for $p = 2$ the two strands are exchanged by one turn around K , so “same strand” is meaningful only relative to a base arc). With $\ell_0(\delta) = 2\tau \sin(\delta/2)$, $E(\delta) = \tau(1 - \cos \delta)$ for $\delta \leq \pi/2$ and $E(\delta) = \tau(1 + \delta - \pi/2)$ for $\delta > \pi/2$, $\beta(\delta) = \delta^2/(4 \sin(\delta/2))$, define

$$\Lambda(\delta) = \ell_0 - \frac{2rE}{\ell_0}, \quad P_1(\delta) = \left[2 \cos\left(\frac{\delta}{2\Theta_{\min}}\right) \sqrt{1 - \beta(\delta)^2} - \frac{\delta^2}{3} \right]_+ \quad (7)$$

(with $P_1 := 0$ if $\delta > \pi\Theta_{\min}$ or $\beta > 1$), and

$$c_1(f, \Theta_{\min}) := \frac{1}{2r} \inf_{0 < \delta < \pi} \max \left\{ \sqrt{\Lambda(\delta)_+^2 + r^2 P_1(\delta)^2}, \ell_0(\delta) - 2r \right\}. \quad (8)$$

All lengths in (8) scale with r , so c_1 depends only on $(f, \Theta_{\min})$; $\ell_0/r - 2$ and $\Lambda/r = (2/f) \sin(\delta/2) - (E/\tau)/\sin(\delta/2)$ decrease in f and P_1 does not depend on f and increases in $\Theta_{\min}$, so c_1 is non-increasing in f and non-decreasing in $\Theta_{\min}$.

Lemma 3.7 (pairs on different strands, $p = 2$). *[proved here] Let $p = 2$, q odd, $r \leq \tau/2$. Any two points X_1, X_2 of K_d with distinct base points and strand index $k = 1$ relative to the shorter base arc satisfy*

$$|X_2 - X_1| \geq \min(2(\tau - r), 2c_1(f, \Theta_{\min})r).$$

Moreover $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$ (a grid evaluation gives 0.6117), hence $c_1 \geq \frac{1}{2}$ whenever $f \leq \frac{1}{2}$ and $\Theta_{\min} \geq \frac{2}{3}$. For the default family $(p, q) = (2, 3)$ with $r_j \leq \text{thick}(K_{j-1})/2$ for all $j \leq d$ one has $\Theta_{\min} \geq \frac{2}{3}$ at every depth, so every such pair is at distance $\geq \min(2(\tau - r), r)$.

Proof. Step 1 (chord of a thick curve). Let $x_0 = K(0)$ and $g(\sigma) = |K(\sigma) - x_0|$ in arclength. A local minimum y_0 of g in $(0, L)$ has $(y_0 - x_0) \perp T(y_0)$, hence $\rho_{pt}(y_0, x_0) = |x_0 - y_0|/2 \geq \tau$, so $|x_0 - y_0| \geq 2\tau$. Consequently $\{g < 2\tau\}$ consists of an initial interval $[0, a)$ and a final one $(L - b, L)$ on which g is monotone (a non-monotone stretch below 2τ would contain an interior local minimum below 2τ), and on them $g' = \cos \angle(T, K - x_0) \geq \sqrt{1 - g^2/4\tau^2}$ a.e. (from $\rho_{pt}(K(\sigma), x_0) \geq \tau$), which integrates to $g(\sigma) \geq 2\tau \sin(\sigma/2\tau)$ and shows $a, b \leq \pi\tau$ (this is the classical chord-arc inequality for thick curves, cf. [3, 7]). Hence for base points with shorter arc $\sigma = \tau\delta \leq L/2$: either $\delta \geq \pi$ and $|x_0 - y_0| \geq 2\tau$ (then Proposition 3.6(b) gives $|X_2 - X_1| \geq 2(\tau - r)$), or $\delta < \pi$ and $\ell := |x_0 - y_0| \geq \ell_0(\delta)$.

Step 2 (elementary estimates, $\delta < \pi$). Let $x_0 = K(t_1)$, $y_0 = K(t_2)$, $e = (y_0 - x_0)/\ell$ and $\bar{T} = \sigma^{-1} \int vT dt$ over the arc, so $e = \bar{T}/|\bar{T}|$, $|\bar{T}| = \ell/\sigma$. Since $|dT/d\sigma| = \kappa \leq 1/\tau$: (E2) $|\bar{T} - T(x_0)| \leq \delta/2$; (E3) for $w \perp T(x_0)$, $|w \cdot e| = |w \cdot \Pi e| \leq |w| |\bar{T} - T(x_0)| \sigma/\ell \leq |w| \beta(\delta)$, where Π is the projection onto the normal plane at x_0 ; (E4) for unit $w \perp T(x_0)$, $|w \cdot (y_0 - x_0)| = |\int v w \cdot \Pi T dt| \leq \int_0^\sigma \sin \min(\sigma'/\tau, \pi/2) d\sigma' = E(\delta)$, and likewise for $w \perp T(y_0)$; (E5) if $\tilde{U}_2$ denotes the vector with the same parallel-frame angle as U_2 but in the normal plane at x_0 , then $U_2 - \tilde{U}_2 = -\int \kappa_\perp vT dt$ with $|\kappa_\perp| \leq \kappa$ (differentiate $\cos \psi_2 N_0 + \sin \psi_2 B_0$ along the arc), so $|U_2 - \tilde{U}_2| \leq \delta$ and its component orthogonal to e is at most $\int \kappa v |T - \bar{T}| dt \leq \int_0^\sigma \frac{\sigma'^2 + (\sigma - \sigma')^2}{2\sigma\tau^2} d\sigma' = \delta^2/3$; (E6) $\tilde{U}_2 - U_1$ lies in the normal plane at x_0 and has length $2|\sin(\Delta\psi/2)|$, where $\Delta\psi = m\sigma + 2\pi qk/p$ is the relative angle of the strands after parallel transport along the arc (the holonomy terms cancel because $N_0(t + 2\pi)$ is $N_0(t)$ rotated by α and m contains $-\alpha/2\pi$), i.e. $|\sin(\Delta\psi/2)| = |\cos(u/2)|$ for $k = 1$ and $p = 2$.

Step 3 (decomposition). Write $X_2 - X_1 = (y_0 - x_0) + r(U_2 - U_1) = A_e e + Q$ with $Q \perp e$. By (E4), $A_e = \ell + r(U_2 - U_1) \cdot e \geq \ell - 2rE/\ell \geq \Lambda(\delta)$, since $\ell \mapsto \ell - 2rE/\ell$ is increasing and $\ell \geq \ell_0$. By (E3), (E5), (E6), $|Q| = r|(U_2 - U_1)_\perp| \geq r[2|\cos(u/2)|\sqrt{1 - \beta^2 - \delta^2/3}]_+$. Finally $u = m|s| \leq m\sigma/v_{\min} = \delta/\Theta_{\min}$ and $|\cos(u/2)|$ is decreasing on $[0, \pi]$, so $|\cos(u/2)| \geq \cos(\delta/2\Theta_{\min})$ when $\delta \leq \pi\Theta_{\min}$; together with the trivial bound $|X_2 - X_1| \geq \ell - 2r \geq \ell_0 - 2r$ this gives $|X_2 - X_1| \geq 2c_1(f, \Theta_{\min})r$ for every pair with $\delta < \pi$.

Step 4 ($c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$). Put $r = 1$, $\tau = 2$, $\Theta_{\min} = 2/3$; we show that the maximum in (8) is ≥ 1 for every $\delta \in (0, \pi)$. For $\delta \in [\pi/3, \pi/2]$, $\Lambda = 2(\tau - r) \sin(\delta/2) = 2 \sin(\delta/2) \geq 1$. For $\delta \in [\pi/2, \pi]$, with $\varsigma = \sin(\delta/2) \in [1/\sqrt{2}, 1]$, $\Lambda = 4\varsigma - (1 + 2 \arcsin \varsigma - \pi/2)/\varsigma$, and $\Lambda \geq 1$ amounts to $\phi(\varsigma) := 4\varsigma^2 - \varsigma - 1 + \pi/2 - 2 \arcsin \varsigma \geq 0$; $\phi'' = 8 - 2\varsigma(1 - \varsigma^2)^{-3/2}$ changes sign once on the interval, so ϕ' is increasing then decreasing, and since $\phi'(1/\sqrt{2}) > 0$ and $\phi'(1^-) = -\infty$, ϕ increases then decreases; its minimum is at an endpoint, $\phi(1/\sqrt{2}) = 0.29$, $\phi(1) = 0.43$. For $\delta \in (0, \pi/3)$ we need $4 \sin^2(\delta/2) + P_1^2 \geq 1$. On $[0.6, \pi/3]$: $4 \sin^2(\delta/2) \geq 4 \sin^2(0.3) > 0.349$, $\cos(3\delta/4) \geq \cos(\pi/4) > 0.7071$, $\beta \leq \beta(\pi/3) < 0.5484$ (β is increasing), so $\sqrt{1 - \beta^2} > 0.8362$, and $\delta^2/3 \leq 0.3656$; hence $P_1 > 2 \cdot 0.7071 \cdot 0.8362 - 0.3656 > 0.8169$, $P_1^2 > 0.667$, and the sum exceeds 1.01. On $(0, 0.6]$: $\cos(3\delta/4) \geq \cos(0.45) \geq 0.900$, $\beta \leq \beta(0.6) \leq 0.305$, $\delta^2/3 \leq 0.12$, so $P_1 \geq 2 \cdot 0.900 \cdot 0.952 - 0.12 \geq 1.59$. All inequalities are non-strict where needed: at $\delta = \pi/3$, $\Lambda = 1$ exactly. Hence $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$. The value 0.6117 (attained near $\delta = 1.23$; an independent evaluation during the review agreed to four digits) is a grid infimum with a Lipschitz constant estimated on the grid itself (`experiments/check_Hc.py`), not a certified bound.

Step 5 (the family). $\Theta_{\min} = v_{\min}/(m\tau)$. By Proposition 3.5, $\tau_{d-1} = \text{thick}(K_{d-1}) \leq r_{d-1} \leq \tau_{d-2}/2$, so $\tau_{d-1} \leq 2^{-(d-1)}$. After the reparametrisation $t \mapsto t/p$, (5) gives $v_{d,\min} \geq p(1 - f)v_{d-1,\min} \geq v_{d-1,\min}$ for $p = 2$, $f \leq 1/2$, so $v_{\min} \geq v_0 = 1$ at every depth; and $m \leq q/p + 1/2 = 2$. Hence $\Theta_{\min} \geq 2^{d-2} \geq 1$ for $d \geq 2$, while for $d = 1$ ($\alpha = 0$, $m = 3/2$, $v = \tau = 1$) $\Theta_{\min} = 2/3$. Monotonicity of c_1 concludes. $\square$

Proposition 3.8 (curvature of K_d). *[proved here, under (H₃)] Assume in addition (H₃): $\sup |v'| \leq M_v$ and $\sup \sqrt{\kappa_1'^2 + \kappa_2'^2} \leq M_\kappa$ for K (derivatives in t , parallel-frame curvatures; $\kappa_1'^2 + \kappa_2'^2 = \kappa'^2 + v^2 \kappa^2 \tau_{\text{tors}}^2$ includes the torsion). Then, with $g(\varsigma) = \varsigma(\varsigma^2 + 2)/(\varsigma^2 + 1)^{3/2}$ and $G(\zeta_{\min}) = \sup_{\varsigma \geq \zeta_{\min}} g \leq g(\sqrt{2}) < 1.0887$: (a)*

$$\kappa_{K_d} \leq \bar{\kappa}_d := \frac{G(\zeta_{\min})}{\tau - r} + \frac{1}{r(1 + \zeta_{\min}^2)} + \frac{rm[M_v(1 + f) + r v_{\max} M_\kappa]}{(v_{\min}^2(1 - f)^2 + r^2 m^2)^{3/2}}, \quad \min \text{Rad}(K_d) \geq 1/\bar{\kappa}_d; \quad (9)$$

(b) $\min \text{Rad}(K_d) \geq 1/\hat{\kappa}_d$, with $A = v^2(1-x)^2$, $B = r^2m^2$, $b_U = v^2(1-x)x/r - rm^2$ and

$$\hat{\kappa}_d := \sup_{|x| \leq f, v_{\min} \leq v \leq v_{\max}} \frac{[b_U^2(A+B) + (rm(M_v(1-x) + rvM_\kappa) + v(A+2B)\sqrt{\tau^{-2} - x^2/r^2})^2]^{1/2}}{(A+B)^{3/2}}.$$

Proof. Let $U = \cos \psi N_0 + \sin \psi B_0$, $W = T \times U$, $\kappa_\perp = \kappa_1 \cos \psi + \kappa_2 \sin \psi$, $\kappa_W = -\kappa_1 \sin \psi + \kappa_2 \cos \psi$ ($\kappa_\perp^2 + \kappa_W^2 = \kappa^2$), $x = r\kappa_\perp \in [-f, f]$. Differentiating (5) with $U' = mW - \kappa_\perp vT$, $W' = -mU - \kappa_W vT$, $T' = v(\kappa_\perp U + \kappa_W W)$:

$$\begin{aligned} K_d' &= c_T T + c_W W, & K_d'' &= aT + b_U U + b_W W, \\ c_T &= v(1-x), & c_W &= rm, & b_U &= v^2(1-x)\kappa_\perp - rm^2, & b_W &= v^2(1-x)\kappa_W, \end{aligned}$$

$a = a_0 - 2rv m \kappa_W$, $a_0 = v'(1-x) - rv(\kappa_1' \cos \psi + \kappa_2' \sin \psi)$. Since (T, U, W) is orthonormal and right-handed, $|K_d' \times K_d''|^2 = b_U^2 |K_d'|^2 + (c_W a - c_T b_W)^2$. Put $A = v^2(1-x)^2$, $B = r^2m^2$, so $|K_d'|^2 = A + B$ and $c_W a - c_T b_W = rma_0 - \kappa_W v(A + 2B)$. By Minkowski, $\kappa_{K_d} \leq \sqrt{v^4(1-x)^2 \kappa_\perp^2 (A+B) + \kappa_W^2 v^2 (A+2B)^2} / (A+B)^{3/2} + (rm^2(A+B)^{1/2} + rm|a_0|) / (A+B)^{3/2}$. In the first term $A(A+B) \leq (A+2B)^2$ and $\kappa_\perp^2 + \kappa_W^2 = \kappa^2$ give $\leq v\kappa(A+2B)/(A+B)^{3/2}$, which is decreasing in $A \geq v^2(1-f)^2$; substituting $A = v^2(1-f)^2$ and $\zeta = v(1-f)/(rm)$ yields $\kappa g(\zeta)/(1-f) \leq G(\zeta_{\min})/(\tau(1-f))$. The second term is at most $1/(r(1+\zeta_{\min}^2)) + rm[M_v(1+f) + rv_{\max}M_\kappa]/(v_{\min}^2(1-f)^2 + r^2m^2)^{3/2}$. (b) By the identity above, $\kappa_{K_d}^2 = [b_U^2(A+B) + (rma_0 - \kappa_W v(A+2B))^2]/(A+B)^3$ with $x = r\kappa_\perp$, $|a_0| \leq M_v(1-x) + rvM_\kappa$ and $|\kappa_W| \leq (\tau^{-2} - x^2/r^2)^{1/2}$; for fixed (x, v) this is non-decreasing in $|a_0|$ and $|\kappa_W|$. $\square$

Same-strand pairs by a turning argument (added in v0.3). For a pair on the same strand ($k = 0$) Step 3 of Lemma 3.7 gives $|Q| \geq r[2|\sin(u/2)|\sqrt{1-\beta^2-\delta^2/3}]_+$, useless as $u \rightarrow 0$, so short doubly critical pairs must be excluded otherwise. Version 0.2 did so with the curvature bound under (H_3) ($c_0 = 0.32, 0.13, 0.16$ at $f = 1/2$, $d = 1, 2, 3$). The following measures instead the turning of the tangent of K_d per unit base arclength and, for $d \geq 2$, uses no derivative of the base; it supersedes the v0.2 proposition, which is omitted.

Lemma 3.9 (turning exclusion). *[proved here] Let Γ be a C^1 arc from X_1 to X_2 with unit tangent T_Γ , T_i its tangent at X_i , $a(\sigma) = \angle(T_\Gamma(\sigma), T_1)$ and $b(\sigma) = \angle(T_\Gamma(\sigma), T_2)$. If $a + b < \pi$ along Γ , then $(X_2 - X_1) \cdot (T_1 + T_2) > 0$; in particular (X_1, X_2) is not doubly critical. This holds whenever Γ is $C^{1,1}$ and its total turning $\int_\Gamma |T_\Gamma'|$ is $< \pi$.*

Proof. As $a, b \geq 0$ and $a + b < \pi$, $\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2} > 0$, so $(X_2 - X_1) \cdot (T_1 + T_2) = \int_\Gamma T_\Gamma \cdot (T_1 + T_2) d\sigma > 0$, while a doubly critical pair has $(X_2 - X_1) \cdot T_1 = (X_2 - X_1) \cdot T_2 = 0$. The last sentence follows from $a(\sigma) \leq \int_0^\sigma |T_\Gamma'|$ and $b(\sigma) \leq \int_\sigma^\ell |T_\Gamma'|$ (T_Γ is Lipschitz, hence absolutely continuous). $\square$

In the notation of the proof of Proposition 3.8, $K_d' = v(1-x)T + rmW$ with $x = r\kappa_\perp \in [-f, f]$, so the angle $\chi \in (0, \pi/2)$ between K_d' and T has $\tan \chi = rm/(v(1-x))$ and $\chi_{\text{lo}} := \arctan \frac{rm}{v_{\max}(1+f)} \leq \chi \leq \chi_{\text{hi}} := \arctan \frac{rm}{v_{\min}(1-f)} = \arctan \frac{1}{\zeta_{\min}}$.

Proposition 3.10 (pairs on the same strand, $p = 2$, sharpened). *[proved here] Let $p = 2$, q odd, $f = r/\tau \leq \frac{1}{2}$, $m > 0$. Set $\delta_{R2} := (\pi - 2(\chi_{\text{hi}} - \chi_{\text{lo}}))/(1 + \sin \chi_{\text{hi}}/\Theta_{\min})$; if (H_3) of Proposition 3.8 holds, set $\delta_{R1} := \pi/(\tau\bar{\Omega})$ with*

$$\tau\bar{\Omega} := 1 + \sin^2 \chi_{\text{hi}} + \frac{\sin \chi_{\text{hi}}}{\Theta_{\min}} + \tau \frac{rm[M_v(1+f)/v_{\min} + rM_\kappa]}{v_{\min}^2(1-f)^2 + r^2m^2},$$

and otherwise $\delta_{R1} := 0$; let $\delta_{\text{turn}} = \max(\delta_{R1}, \delta_{R2})$. Then every doubly critical pair of K_d with distinct base points and strand index $k = 0$ relative to the shorter base arc has $\delta \geq \delta_{\text{turn}}$ and

$$|X_2 - X_1| \geq \min(2(\tau - r), 2c'_0 r), \quad c'_0 := \frac{1}{2r} \inf_{\delta_{\text{turn}} \leq \delta < \pi} \max(\Lambda(\delta), \ell_0(\delta) - 2r).$$

In particular $c'_0 \geq \frac{1}{2}$ whenever $\delta_{\text{turn}} \geq \pi/3$.

Proof. If $\delta \geq \pi$ or the base chord is $\geq 2\tau$, Step 1 of Lemma 3.7 and Proposition 3.6(b) give $|X_2 - X_1| \geq 2(\tau - r)$. Otherwise let Γ be the arc of K_d over the shorter base arc, $t \in [t_1, t_2]$, so $u = m(t_2 - t_1) \leq \delta/\Theta_{\min}$; since $k = 0$, Γ joins X_1 to X_2 .

$\delta \geq \delta_{R2}$, without (H_3) . Put $\chi_i = \chi(t_i)$ and $Y_i(t) = \cos \chi_i T(t) + \sin \chi_i W(t)$. From $U' = mW - v\kappa_\perp T$, $W' = -mU - v\kappa_W T$, $T' = v(\kappa_\perp U + \kappa_W W)$ we get $Y'_i = (v\kappa_\perp \cos \chi_i - m \sin \chi_i)U + v\kappa_W \cos \chi_i W - v\kappa_W \sin \chi_i T$ and $|Y'_i|^2 = (v\kappa_\perp \cos \chi_i - m \sin \chi_i)^2 + v^2 \kappa_W^2 \leq (v\kappa + m \sin \chi_i)^2$, using $\kappa_\perp^2 \cos^2 \chi_i + \kappa_W^2 \leq \kappa^2$ and $-\kappa_\perp \cos \chi_i \leq \kappa$. The unit tangent $T_d(t) = \cos \chi(t) T(t) + \sin \chi(t) W(t)$ and $Y_i(t)$ lie in $\text{span}(T(t), W(t))$ at angles $\chi(t)$, χ_i from $T(t)$, and $Y_i(t_i) = T_d(t_i)$, so $a(t) \leq |\chi(t) - \chi_1| + \int_{t_1}^t (v\kappa + m \sin \chi_1)$ and $b(t) \leq |\chi(t) - \chi_2| + \int_t^{t_2} (v\kappa + m \sin \chi_2)$. With $\int v\kappa dt \leq \sigma/\tau = \delta$, $a + b \leq 2(\chi_{hi} - \chi_{lo}) + \delta + u \sin \chi_{hi} \leq 2(\chi_{hi} - \chi_{lo}) + \delta(1 + \sin \chi_{hi}/\Theta_{\min})$, which is $< \pi$ if $\delta < \delta_{R2}$; Lemma 3.9 excludes the pair.

$\delta \geq \delta_{R1}$, under (H_3) . The turning rate of T_d per unit t is $\varpi = |K'_d \times K''_d|/|K'_d|^2$. The Minkowski split in the proof of Proposition 3.8, multiplied by $|K'_d| = \sqrt{A+B}$, gives $\varpi \leq v\kappa(A+2B)/(A+B) + rm^2/\sqrt{A+B} + rm|a_0|/(A+B)$, where $B/(A+B) = \sin^2 \chi$ and $rm/\sqrt{A+B} = \sin \chi$; with $\kappa \leq 1/\tau$, $\chi \leq \chi_{hi}$, $v \geq v_{\min}$ and $|a_0| \leq M_v(1+f) + rvm_\kappa$ this is $\varpi/v \leq \bar{\Omega}$ per unit base arclength, so the total turning of Γ is $\leq \tau \bar{\Omega} \delta$, which is $< \pi$ if $\delta < \delta_{R1}$.

Distance. For $\delta < \pi$, Step 3 of Lemma 3.7, which does not use the strand index, gives $|X_2 - X_1| \geq \Lambda(\delta)$ and $|X_2 - X_1| \geq \ell_0(\delta) - 2r$; hence c'_0 . Finally $\Lambda \geq r$ on $[\pi/3, \pi)$ when $f \leq \frac{1}{2}$: on $[\pi/3, \pi/2]$, $\Lambda = 2(\tau - r) \sin(\delta/2) \geq \tau - r \geq r$; on $[\pi/2, \pi)$ this is Step 4 of Lemma 3.7, and Λ/r decreases in f . $\square$

Corollary 3.11 (same strand in the default family). *[proved here] Let $(p_j, q_j) = (2, 3)$ and $f_j = r_j/\text{thick}(K_{j-1}) \leq \frac{1}{2}$ for all $j \leq d$. Then every doubly critical pair of K_d with distinct base points on the same strand is at distance $\geq \min(2(\tau - r), 2c'_0 r)$ with $c'_0 \geq \frac{1}{2}$, at every depth $d \geq 1$, without (H_3) . At $d = 1$ and $f = \frac{1}{2}$, $\delta_{\text{turn}} \geq \delta_{R1} = \pi/(1 + \frac{9}{13} + \frac{9}{2\sqrt{13}}) > 1.0684$ and $c'_0 \geq \sin(\delta_{R1}/2) > 0.509$ (and $c'_0 > 0.509$ for every $f \leq \frac{1}{2}$); for $d \geq 2$, $\delta_{\text{turn}} \geq \delta_{R2} > 1.677$.*

Proof. $d = 1$. The base is the unit circle: $v \equiv 1$, $\kappa \equiv 1 = 1/\tau$, $\alpha = 0$, $m = \frac{3}{2}$, $\Theta_{\min} = \frac{2}{3}$, and (H_3) holds with $M_v = M_\kappa = 0$. With $\zeta = \zeta_{\min} = 2(1-f)/(3f)$, $\tau\bar{\Omega} = 1 + \frac{1}{1+\zeta^2} + \frac{3}{2\sqrt{1+\zeta^2}}$, which is $< 2.9404 < 3$

at $f = \frac{1}{2}$ ($\zeta = \frac{2}{3}$) and decreases as f decreases; so $\delta_{R1}(f) \geq \delta_{R1}(\frac{1}{2}) > \pi/3$. We argue at $f = \frac{1}{2}$, with $c_* := 2 \sin(\delta_{R1}(\frac{1}{2})/2)$ fixed. On $[\delta_{R1}(\frac{1}{2}), \pi/2]$, $\Lambda/r = 2(1/f - 1) \sin(\delta/2) \geq c_*$. On $[\pi/2, \pi)$ the argument of Step 4 of Lemma 3.7 with 1 replaced by c_* goes through: $\phi_{c_*}(\zeta) = 4\zeta^2 - c_*\zeta - 1 + \pi/2 - 2 \arcsin \zeta$ has the same second derivative, $\phi'_{c_*}(1/\sqrt{2}) = 2\sqrt{2} - c_* > 1.81$, $\phi_{c_*}(1/\sqrt{2}) = 1 - c_*/\sqrt{2} > 0.279$ and $\phi_{c_*}(1) = 3 - c_* - \pi/2 > 0.410$; so $c'_0(\frac{1}{2}) \geq c_*/2 > 0.509$. For $f < \frac{1}{2}$ the interval $[\delta_{\text{turn}}(f), \pi)$ is contained in $[\delta_{R1}(\frac{1}{2}), \pi)$ and Λ/r increases as f decreases, so $c'_0(f) \geq c_*/2$ as well.

$d \geq 2$. By (5), $|K'_1|^2 = (1-x)^2 + r_1^2 m_1^2$ with $r_1 = f_1$, $m_1 = \frac{3}{2}$, so after the reparametrisation $v_{\min}(K_1) \geq 2\sqrt{(1-f_1)^2 + \frac{9}{4}f_1^2}$, while $\text{thick}(K_1) \leq f_1$ (Proposition 3.5); hence $v_{\min}(K_1)/\text{thick}(K_1) \geq 2\sqrt{(1-f_1)^2/f_1^2 + \frac{9}{4}} \geq \sqrt{13}$. For $j \geq 2$, $v_{\min}(K_j) \geq 2(1-f_j)v_{\min}(K_{j-1})$ and $\text{thick}(K_j) \leq f_j \text{thick}(K_{j-1})$, so $v_{\min}/\text{thick}$ at least doubles per depth. With $m_d \leq \frac{3}{2} + \frac{1}{2} = 2$, $\Theta_{\min} \geq \sqrt{13}2^{d-3} \geq \sqrt{13}/2 > 1.802$ and $\zeta_{\min} = (1-f_d)\Theta_{\min}/f_d \geq \Theta_{\min}$. Dropping $\chi_{lo} \geq 0$, $\delta_{R2} \geq (\pi - 2 \arctan(1/\zeta_{\min}))/((1+1)/(\Theta_{\min}\sqrt{1+\zeta_{\min}^2}))$, which increases in $\Theta_{\min}$ and $\zeta_{\min}$ and at $\Theta_{\min} = \zeta_{\min} = \sqrt{13}/2$ exceeds $1.677 > \pi/3$. (The bound $\Theta_{\min} \geq 2^{d-2}$ of Step 5 of Lemma 3.7 would not suffice at $d = 2$: at $f = \frac{1}{2}$, $\delta_{R2} \geq \pi/3$ needs $\Theta_{\min} \geq 1.112$.) Proposition 3.10 concludes. $\square$

Theorem 3.12 (thickness lower bound for $p = 2$). *[proved here; (b) under (H_3) ; the values in Table 1 are grid evaluations, not certified] Let $p = 2$, q odd, $f = r/\tau \leq 1/2$, $m > 0$. (a) $\frac{1}{2} \text{dcsd}(K_d) \geq \min(\tau - r, \min(c_1, c'_0)r)$, with $c_1 = c_1(f, \Theta_{\min})$ of Lemma 3.7 and c'_0 of Proposition 3.10. (b) Under (H_3) , $\min \text{Rad}(K_d) \geq 1/\hat{\kappa}_d$, hence $\text{thick}(K_d) \geq \min(\tau - r, cr)$ with $c = \min(c_1, c'_0, 1/(r\hat{\kappa}_d))$. (c) For the default family $(2, 3)$ with $f_j \leq \frac{1}{2}$ at every level, $\frac{1}{2} \text{dcsd}(K_d) \geq \min(\tau - r, r/2) = r/2$ and $\text{thick}(K_d) \geq \min(\min \text{Rad}(K_d), r/2)$ at every depth, without (H_3) ; at $d = 1$, $\text{thick}(K_1) \geq r_1/2$; for $d \geq 2$, (H_c) with $c = \frac{1}{2}$ holds at level d if and only if $\min \text{Rad}(K_d) \geq r_d/2$.*

Proof. By (1), $\text{thick} = \min(\min \text{Rad}, \frac{1}{2} \text{dcsd})$. Doubly critical pairs with the same base point are at distance $2r$ (Proposition 3.6(a)); if the shorter base arc has $\delta \geq \pi$, the base chord is $\geq 2\tau$ (Step 1 of Lemma 3.7) and the distance is $\geq 2(\tau - r)$ (Proposition 3.6(b)); pairs with $\delta < \pi$ are covered by Lemma 3.7 ($k = 1$) and Proposition 3.10 ($k = 0$). Since $c_1 \leq 1$ (its limit as $\delta \rightarrow 0$ is 1), this gives (a); (b) is Proposition 3.8(b). For (c), $c_1 \geq \frac{1}{2}$ is Lemma 3.7, $c'_0 \geq \frac{1}{2}$ is Corollary 3.11, and $\tau - r \geq r$. At $d = 1$ ($M_v = M_\kappa = 0$, $v \equiv 1$), Proposition 3.8 gives $r\hat{\kappa}_1 = fG(\zeta_{\min})/(1-f) + 1/(1+\zeta_{\min}^2)$ with $\zeta_{\min} = 2(1-f)/(3f)$ and $G \leq \max g = g(\sqrt{2}) = 4\sqrt{2}/(3\sqrt{3})$ (the derivative of $y(y+2)^2/(y+1)^3$, $y = \zeta^2$, is proportional to $(y+2)(2-y)$); this increases with f , and at $f = \frac{1}{2}$ equals $g(\sqrt{2}) + \frac{9}{13}$, so $1/(r\hat{\kappa}_1) > 0.561 > \frac{1}{2}$. The equivalence for $d \geq 2$ follows from $\text{thick} = \min(\min \text{Rad}, \frac{1}{2} \text{dcsd})$, $\frac{1}{2} \text{dcsd} \geq r/2$ and $\min(\tau - r, \frac{r}{2}) = \frac{r}{2}$. $\square$

In Table 1, $c \geq 0.509$ in every measured level, and the minimum is the different-strand constant c_1 except at $(f = \frac{1}{2}, d = 1)$, where it is c'_0 .

Table 1: Constants of Theorem 3.12 on the polygons ($N_0 = 256$; τ, v, α and, for $1/(r\hat{\kappa}_d)$ at $d \geq 2$, M_v, M_κ measured on the base polygon; $\delta_{\text{turn}} = \delta_{R1}$ at $d = 1$, δ_{R2} at $d \geq 2$), rounded down. Grid evaluations, not certified (`experiments/check_Hc.py`, `check_same_strand.py`); analytic: $c_1, c'_0 \geq \frac{1}{2}$ in every row. At $f = 0.25$ all three are ≥ 0.999 . Last column: measured τ_d/r_d . At $d = 2$ the column $1/(r\hat{\kappa}_d)$ is superseded by the certified $\min\text{Rad}(K_2) \geq 0.842 r_2$ (Proposition 3.19).

f	d	c_1	c'_0 (δ_{turn})	$1/(r\hat{\kappa}_d)$	c	τ_d/r_d
0.35	1	0.965	1.156 (1.343)	2.215	0.965	1.000
0.35	2	0.999	1.824 (2.838)	1.878	0.999	1.000
0.35	3	0.999	1.856 (3.107)	1.855	0.999	1.000
0.5	1	0.611	0.509 (1.068)	1.419	0.509	0.832
0.5	2	0.699	0.845 (2.056)	1.034	0.699	1.000
0.5	3	0.706	0.988 (2.928)	0.998	0.706	1.000

Remark 3.13 (no constant works for an arbitrary base). [proved here] No $c > 0$ depending only on (p, q, f) makes $\text{thick}(K_d) \geq \min(\tau - r, cr \sin(\pi/p))$ true for every smooth base K with $\text{thick}(K) = \tau$. (i) If the curvature of K rises from 0 to $1/\tau$ over an arc of length ε (a smoothed junction of a segment and a circle of radius τ), by (5) the tangent of the inner strand turns from $vT + rmW$ to $v(1-f)T + rmW$ over that arc, so $\kappa_{K_d} \gtrsim rv\kappa'/|K'_d|^2 \rightarrow \infty$: the M_v, M_κ term of (9) is unavoidable. (ii) If K is traversed slowly ($v \rightarrow 0$) on an arc of length $\gg r$, the two strands there are helices of pitch $h = v/m \rightarrow 0$; in the straight-tube model $|K_d(t) - K_d(t + 2\pi + u)|^2 = 4r^2 \cos^2(qu/4) + v^2 u^2$, whose minimum over u is $\leq (2\pi v/q)^2$ and is attained at a doubly critical pair by screw symmetry, so $\text{thick}(K_d) \leq \pi v/q \rightarrow 0$ (in Lemma 3.7, $c_1 \rightarrow 0$ as $\Theta_{\min} \rightarrow 0$). In the family $\Theta_{\min}$ is bounded below a priori (Lemma 3.7, Corollary 3.11); no a priori bound on M_v, M_κ uniform in d is known (but see Corollary 3.16 and, for $d = 2$, Proposition 3.19).

Towards (H₃) uniformly in d (added in v0.4). For a regular closed curve K with parameter t of period 2π , speed v , curvature vector $\kappa = dT/ds = \kappa_1 N_0 + \kappa_2 B_0$ and $\kappa'_{\text{par}} := \kappa'_1 N_0 + \kappa'_2 B_0$ (the normal part of $d\kappa/dt$), set $\nu(K) := \sup_t |v'|/v^2$ and $\mu(K) := \sup_t |\kappa'_{\text{par}}|/v$. ν is invariant under linear reparametrisations ($\tilde{v}(s) = pv(ps)$ gives $\tilde{v}'/\tilde{v}^2 = v'/v^2$), $\mu = \sup |\Pi_\perp d\kappa/ds|$ is geometric, and $M_v \leq \nu v_{\max}^2$, $M_\kappa \leq \mu v_{\max}$. In the notation of the proof of Proposition 3.8 put $\varepsilon := r\kappa_{\max}(K) \leq f$, $\lambda := rm/v_{\min}$ and $\zeta' := v_{\min}(1 - \varepsilon)/(rm) \geq \zeta_{\min}$. The four statements below were derived in `theory/` after the second review, checked by the author (symbolic frame calculus and a smooth computation, `experiments/verify_H3.py`) and by an independent internal referee (round 4).

Lemma 3.14 (speed of K_d). [proved here; checked by an independent internal referee (round 4)] With $w = |K'_d|$, $c_T = v(1 - x)$ and $a_0 = v'(1 - x) - rv\kappa'_{\text{par}} \cdot U$,

$$w' = \frac{c_T(a_0 - rvm\kappa_W)}{w}, \quad \text{hence} \quad \nu(K_d) \leq \frac{\nu(K)}{1 - \varepsilon} + \frac{r\mu(K) + \lambda\kappa_{\max}(K)}{(1 - \varepsilon)^2}. \quad (10)$$

Proof. $w^2 = c_T^2 + r^2 m^2$ with rm constant, so $w' = c_T c'_T/w$, and $c'_T = v'(1 - x) - vx'$ with $x' = r(\kappa'_{\text{par}} \cdot U + m\kappa_W)$ (as $\kappa \cdot U' = m\kappa_W$). Then $|w'|/w^2 \leq |a_0 - rvm\kappa_W|/c_T^2$ (since $w \geq c_T$), and $|v'| \leq \nu v^2$, $|\kappa'_{\text{par}}| \leq \mu v$, $|\kappa_W| \leq \kappa_{\max}$, $1 - x \geq 1 - \varepsilon$, $v \geq v_{\min}$ give the bound; $\nu(K_d)$ may be computed in any linear reparametrisation. $\square$

Proposition 3.15 (curvature of K_d from the data of the base). [proved here; checked by an independent internal referee (round 4)] Under the hypotheses of Proposition 3.8, with G as there,

$$\kappa_{\max}(K_d) \leq \frac{\kappa_{\max}(K) G(\zeta')}{1 - \varepsilon} + \frac{1}{r(1 + \zeta'^2)} + \frac{\lambda(\nu(K)(1 + \varepsilon) + r\mu(K))}{(1 - \varepsilon)^3}. \quad (11)$$

Proof. The Minkowski split of the proof of Proposition 3.8(a) with $|x| \leq \varepsilon$ and $\kappa \leq \kappa_{\max}(K)$: the first term is $\leq v\kappa(A + 2B)/(A + B)^{3/2}$, decreasing in $A \geq v^2(1 - \varepsilon)^2$, hence $\leq \kappa_{\max} G(\zeta')/(1 - \varepsilon)$; $rm^2/(A + B) \leq 1/(r(1 + \zeta'^2))$; and $rm|a_0|/(A + B)^{3/2} \leq rm v^2(\nu(1 + \varepsilon) + r\mu)/(v^3(1 - \varepsilon)^3) \leq \lambda(\nu(1 + \varepsilon) + r\mu)/(1 - \varepsilon)^3$. $\square$

Corollary 3.16 (tolerated growth of the (H_3) data, default family). *[proved here; checked by an independent internal referee (round 4)]* Let $(p_j, q_j) = (2, 3)$ and $r_j \leq f \text{ thick}(K_{j-1})$, $f \leq \frac{1}{2}$, for all $j \leq d$, and let $d \geq 3$. If

$$\frac{96}{\sqrt{13}} 4^{-d} \nu(K_{d-1}) + \frac{64}{\sqrt{13}} 8^{-d} \mu(K_{d-1}) \leq 2 - g(\sqrt{13}) - \frac{1}{14} \quad (= 0.8961 \dots), \quad (12)$$

then $\min\text{Rad}(K_d) \geq r_d/2$, hence $\text{thick}(K_d) \geq r_d/2$: (13) below holds at depth d with $c = \frac{1}{2}$. (Each term may take the whole right-hand side only if the other vanishes: $\nu(K_{d-1}) \leq 0.0336 \cdot 4^d$ with $\mu = 0$, or $\mu(K_{d-1}) \leq 0.0504 \cdot 8^d$ with $\nu = 0$; in general the joint condition (12) applies.)

Proof. By the proof of Corollary 3.11, $\zeta' \geq \zeta_{\min} \geq \Theta_{\min} \geq \sqrt{13} 2^{d-3} \geq \sqrt{13}$, so $G(\zeta') \leq g(\sqrt{13})$ (g decreases on $[\sqrt{2}, \infty)$) and $1/(1 + \zeta'^2) \leq 1/14$; $\varepsilon \leq \frac{1}{2}$ gives $\varepsilon/(1 - \varepsilon) \leq 1$, $(1 + \varepsilon)/(1 - \varepsilon)^3 \leq 12$, $(1 - \varepsilon)^{-3} \leq 8$; Proposition 3.5 gives $r_d \leq 2^{-d}$; and $\lambda = (1 - f_d)/\zeta_{\min} \leq 2^{3-d}/\sqrt{13}$. Multiplying (11) by r_d , $r_d \kappa_{\max}(K_d) \leq g(\sqrt{13}) + \frac{1}{14} + 12r_d \lambda \nu + 8r_d^2 \lambda \mu \leq 2$ under (12); Theorem 3.12(c) concludes. $\square$

Proposition 3.17 (the recursion for (H_3) does not close). *[proved here; checked by an independent internal referee (round 4)]* Fourth derivatives of K enter $K_d'''' = K'''' + rU''''$ only through the term $-rv(\kappa_{\text{par}}'' \cdot U)T$ of rU'''' : with $n_\chi := \sin \chi T - \cos \chi W$ the unit normal of K_d in $\text{span}(T, W)$ and $\sin \chi = rm/w > 0$, the normal part of K_d'''' is $-rv \sin \chi (\kappa_{\text{par}}'' \cdot U) n_\chi + R$, with $|R|$ bounded in terms of $v_{\max}, \kappa_{\max}, \nu, \mu, \sup |v''|, r, m$. Consequently no locally bounded F satisfies $\mu(K_d) \leq F(v_{\min}, v_{\max}, \kappa_{\max}, \nu, \mu, \sup |v''|, r, m)$ for all smooth bases K : the speed and curvature of K_d are controlled by the (H_3) data of K , but $\mu(K_d)$ is not.

Proof. $U'''' = mW'' - (v\kappa_\perp)''T - 2(v\kappa_\perp)'T' - v\kappa_\perp T''$, where $W'' = -mU' - (v\kappa_W)'T - v\kappa_W T'$ and $T'' = v'\kappa + v d\kappa/dt$ involve at most third derivatives of K , and $\kappa_\perp'' = \kappa_{\text{par}}'' \cdot U + 2m\kappa_{\text{par}}' \cdot W - m^2\kappa_\perp$; the normal part of T is $\sin \chi n_\chi$ (a symbolic computation in the frame (T, U, W) confirms the coefficient; v'' enters R with coefficient $(1 - x) \sin \chi$). For the last claim take a smooth closed K with $r < \text{thick}(K)$ and $K_\eta = K + \eta^{7/2} \Phi((t - t_0)/\eta) U(t_0)$ with Φ smooth, compactly supported, $\Phi''''(0) \neq 0$. For small η the support lies in one period, K_η is regular and $r < \text{thick}(K_\eta)$, so $K_{d,\eta}$ is defined and embedded (thickness is lower semicontinuous under C^2 convergence: $\min\text{Rad}$ is continuous, and doubly critical pairs at distance $\leq 2r$ converge, after extraction, to a doubly critical pair of K , collapsing pairs being excluded by Lemma 3.9). $K_\eta \rightarrow K$ in C^3 , so the arguments of F converge, while $|\kappa_{\text{par}}'' \cdot U|(t_0) \gtrsim \eta^{-1/2}$; since the normal part of K_d'''' is $3ww'\kappa_d + w^3 \Pi_\perp d\kappa_d/d\sigma$ with w, w', κ_{K_d} bounded by Lemma 3.14 and Proposition 3.15, $\mu(K_{d,\eta}) \rightarrow \infty$. $\square$

Remark 3.18 (what this gives and what it does not). [grid evaluation with polygon inputs; uniformity open] Lemma 3.14 and Proposition 3.15 propagate ν and $\kappa_{\max}$ given μ at the previous depth, and Corollary 3.16 reduces (H_c) with $c = \frac{1}{2}$ at depth $d \geq 3$ to a growth condition. This does *not* prove (H_c) for $d \geq 3$: by Proposition 3.17 an a priori bound on $\mu(K_{d-1})$ cannot be obtained from the (H_3) data of K_{d-2} alone; a recursive argument must carry higher derivatives (e.g. all of them, via analyticity) or use the explicit structure of the family (one derivative enters per depth, with coefficient $r_d \sin \chi_d \leq 4.44 \cdot 4^{-d}$ a priori, roughly 8^{-d} on the polygons). Proposition 3.17 says nothing about the family itself, where μ grows slowly (values below). At $d = 2$ the a priori constants do not suffice: the analogue of (12) would need $\frac{6}{\sqrt{13}} \nu(K_1) + \frac{1}{\sqrt{13}} \mu(K_1) \leq 2 - g(\frac{\sqrt{13}}{2}) - \frac{4}{17} < 0.685$, while the polygon values give 1.95; $d = 2$ is settled instead by Proposition 3.19. On the polygons ($f = \frac{1}{2}$, $N_0 = 256$; $d = 4$ at $N_0 = 128$, slightly below the smooth values 1.36, 7.0 of `experiments/smooth_H3.py`): $\nu = 0.42, 0.72, 1.12, 1.35$ and $\mu = 4.5, 5.0, 6.2, 6.9$ at $d = 1, \dots, 4$; the left-hand side of (12) is ≤ 0.48 ($d = 3$) and ≤ 0.15 ($d = 4$; ≤ 0.15 with the smooth data); and (10), (11) hold in all 16 checked levels ($f \in \{0.25, 0.35, 0.5\}$) with bound/measured ≥ 1.01 (`experiments/check_H3.py`).

The case $d = 2$ (added in v0.5). At $d = 2$ the base K_1 is an explicit trigonometric polynomial, so instead of the a priori constants one can insert into (11) rigorous enclosures of the four numbers of K_1 that it uses.

Proposition 3.19 (the case $d = 2$). *[computer-assisted proof (interval arithmetic); not yet checked by an independent referee]* Let $(p_1, q_1) = (p_2, q_2) = (2, 3)$, $0 < r_1 \leq \frac{1}{2}$ and $0 < r_2 \leq r_1/2$, with arbitrary phases φ_1, φ_2 (equivalently, arbitrary initial normals of the frames of K_0 and K_1). Then

$$r_2 \kappa_{\max}(K_2) \leq 1.187, \quad \text{hence} \quad \min\text{Rad}(K_2) \geq 0.842 r_2 > r_2/2.$$

Consequently, for every $f \leq \frac{1}{2}$, every $r_1 \leq f$ and every $r_2 \leq f \text{ thick}(K_1)$, $\text{thick}(K_2) \geq r_2/2$: Hypothesis 3.21 with $c = \frac{1}{2}$ holds at $d = 2$ for every choice of phases. Along the family $r_1 = f$, $r_2 \leq f^2$ one has moreover $\min\text{Rad}(K_2) > r_2$ for every $f \leq 0.484375$.

Proof. Reduction. By Proposition 3.15 with $K = K_1$, $r = r_2$, $r_2\kappa_{\max}(K_2) \leq \Phi$, where Φ is r_2 times the right-hand side of (11), a function of $\kappa_{\max}(K_1)$, $v_{\min}(K_1)$, $\nu(K_1)$, $\mu(K_1)$, r_2 and m_2 ; its proof uses them only through $|\kappa| \leq \kappa_{\max}$, $|v'| \leq \nu v^2$, $|\kappa'_{\text{par}}| \leq \mu v$, $v \geq v_{\min}$, so bounds may be inserted. Φ is non-decreasing in $\kappa_{\max}$, ν , μ , r_2 and m_2 and non-increasing in $v_{\min}$: $\varepsilon = r_2\kappa_{\max}$ and $\lambda = r_2m_2/v_{\min}$ increase, $\zeta' = v_{\min}(1 - \varepsilon)/(r_2m_2)$ decreases, and G is non-increasing. Since $\alpha_2 \in (-\pi, \pi]$, $m_2 = \frac{3}{2} - \alpha_2/2\pi < 2$, so m_2 may be replaced by 2: the Bishop frame of K_1 does not enter. By Proposition 3.5, $\text{thick}(K_1) \leq r_1$, so $r_2 \leq f \text{thick}(K_1)$ with $f \leq \frac{1}{2}$ implies $r_2 \leq r_1/2$. A change of φ_1 moves K_1 by a rigid motion and a shift of the parameter ($K_1^\varphi(s) = R_z(-2\varphi/3)K_1^0(s + \varphi/3)$), which leave the four numbers unchanged; φ_2 enters (11) only through $|U| = 1$.

The base. With $t = 2s$, $K_1(s) = (e^{2is}(1 + r_1 \sin 3s), r_1 \cos 3s)$ (first two coordinates as a complex number; $x + iy = e^{2is} + \frac{r_1}{2i}(e^{5is} - e^{-is})$), with explicit derivatives $D_j = d^j K_1/ds^j$. With $v = |D_1|$: $\kappa = |D_1 \times D_2|/v^3$, $|v'|/v^2 = |D_1 \cdot D_2|/v^3$ and, from $\Pi_\perp K_1''' = 3vv'\kappa + v^2\Pi_\perp \kappa'$ and $\kappa = \Pi_\perp D_2/v^2$, $|\kappa'_{\text{par}}|/v = |\Pi_\perp D_3 - 3(D_1 \cdot D_2)\Pi_\perp D_2/v^2|/v^3$, where $\Pi_\perp$ is the projection orthogonal to D_1 . Moreover $K_1(s + 2\pi/3) = R_z(4\pi/3)K_1(s)$ exactly, so suprema over $[0, 2\pi]$ equal suprema over $[0, 2\pi/3]$.

Enclosure. Cover $[0, \frac{1}{2}] \times [0, 2\pi/3]$ by 512×2048 boxes in (r_1, s) (r_1 -endpoints exact binary fractions, s -endpoints enclosed in `mpmath.iv`); on each box enclose $\cos ks, \sin ks$ ($k = 1, 2, 3, 5$) with `mpmath.iv` (80 bits, rounded outward to binary64), and evaluate the expressions above in binary64 interval arithmetic with every endpoint rounded outward by one ulp after each IEEE-754 operation ($+$, $-$, $\times$, $\div$, $\sqrt{}$), which encloses the exact value. On each r_1 -box $[a, b]$ this gives upper bounds of $\kappa_{\max}$, ν , μ and a lower bound of $v_{\min}$ over the box; Φ evaluated in the same arithmetic at these values with $r_2 = b/2$ and $m_2 = 2$ is at most 1.187 on every box (the maximum is on the last box, $r_1 \in [0.499, 0.5]$). Theorem 3.12(c) gives $\text{thick}(K_2) \geq r_2/2$. For the last claim the same box data with $r_2 = b^2$ give $\Phi < 1$ on every box with $b \leq 0.484375$. $\square$

Remark 3.20 (what is certified, and how). The certificate covers exactly $r_2\kappa_{\max}(K_2) \leq 1.187$ for $r_1 \leq \frac{1}{2}$, $r_2 \leq r_1/2$; the rest is Theorem 3.12(c), proved by hand. A second computation over the full period (4096 boxes, no symmetry) gives at $r_1 = \frac{1}{2}$: $v_{\min}(K_1) \geq 1.798$, $\kappa_{\max}(K_1) \leq 1.427$, $\nu(K_1) \leq 0.433$, $\mu(K_1) \leq 4.638$ and $\Phi \leq 1.202$ ($r_2 \leq \frac{1}{4}$). The trust base is IEEE-754 binary64 arithmetic in NumPy, `mpmath.iv`, and about one hundred lines of code implementing the formulas above (`experiments/certify_d2.py`, a frozen copy of the script of `theory/`; 8s); it is not a formal verification. An independent check of the implementation by the author (`experiments/check_certify_d2.py`: other formulas for ν , μ ; float values on the full period against every box enclosure; 368 640 point-in-box tests; the interval class against exact rationals; Φ in 50 digits) found no violation. The bound is not sharp: (11) takes the four suprema separately and uses $m_2 \leq 2$; the smooth (uncertified) value of $r_2\kappa_{\max}(K_2)$ at $f = \frac{1}{2}$, $r_2 = \frac{1}{4}$ and default phases is 0.40 (`verify_H3.py`) against 1.202. This is why $\min\text{Rad}(K_2) > r_2$ is obtained only for $f \leq 0.484375$. Nothing at $d \geq 3$, and no polygonal quantity, is certified.

What remains unproved is therefore a statement about the family, not about an arbitrary curve:

Hypothesis 3.21 (H_c , for the family). [for (2, 3) and $c = 1/2$: proved at $d = 1$ and (computer-assisted) at $d = 2$; for $d \geq 3$ reduced to a curvature bound (Theorem 3.12) and to a growth condition (Corollary 3.16); verified numerically] Let (K_d) be the family of Definition 2.1 built from the circle with a fixed pattern (p, q) , and let $f \leq 1/2$. There is $c = c(p, q, f) \in (0, 1]$ such that, for every sequence (r_d) with $r_d \leq f \text{thick}(K_{d-1})$ and every $d \geq 1$,

$$\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, c r_d \sin(\pi/p)). \quad (13)$$

Status for (2, 3), $f \leq 1/2$, $c = 1/2$. By Theorem 3.12(c), (13) holds at $d = 1$ and, for $d \geq 2$, is equivalent to $\min\text{Rad}(K_d) \geq r_d/2$. At $d = 2$ this is Proposition 3.19 (computer-assisted; every $f \leq \frac{1}{2}$ and every choice of phases). For $d \geq 3$, under (H_3) it follows from $1/(r\hat{\kappa}_d) \geq 1/2$, which the grid evaluation with polygon-measured inputs gives at $d = 3$ (0.998 at $f = 1/2$; Table 1), and from (12), which the polygons satisfy at $d = 3, 4$ (Remark 3.18). This is evidence, not a proof: ν , μ (and, for $\hat{\kappa}_d$, $\text{thick}(K_{d-1})$) are only measured. Open: (H_3) uniformly in d (here, a priori control of $\mu(K_{d-1})$) and certification for $d \geq 3$. Numerically (13) with $c = 1/2$ holds in all 9 default levels with margin $\tau_d/\rho_d \geq 1.66$, and along the chain of Corollary 3.22 with margin ≥ 1.66 .

On the constant. The natural constant is $c = 1$: $r \sin(\pi/p)$ is half the strand chord, and by Propositions 3.5 and 3.6(a) it cannot be improved where $\tau_d = r_d$. With $c = 1$ the inequality fails on the polygons in 4 of the 15 measured levels with $f \leq 1/2$ ((2, 3), $f = 0.5$, $d = 1$ (0.832); (3, 2), $d = 1$ (0.968); (3, 2), $d = 2$ (0.999); (2, 5), $d = 1$ (0.564); the deficit at (3, 2), $d = 2$ is stable under doubling of N); it holds with equality, to 0.05%, in every level with $\tau_d = r_d$ and in (3, 2), $d = 3$; (2, 5), $d = 2, 3$, where the critical pair

is a pair of adjacent strands of one disc. The value $c = 1/2$ was chosen after $c = 1$ had failed (Section 4); Theorem 3.12 reaches it, while $\tau_1/r_1 = 0.832$ at $f = 1/2$ shows that no argument reaches $c = 1$ there.

Corollary 3.22 (recursion for the ropelength bound). *[proved here; for $(2, 3)$, $c = \frac{1}{2}$, $f \leq \frac{1}{2}$: unconditional for $d \leq 2$ (Proposition 3.19 computer-assisted), conditional on $\min\text{Rad}(K_d) \geq r_d/2$ for $d \geq 3$] Fix (p, q) , $f \leq 1/2$ and assume (H_c) with constant c (for all $r_d \leq f \text{thick}(K_{d-1})$). Define $\rho_0 = 1$, $r_d := f\rho_{d-1}$ and $\rho_d := \gamma\rho_{d-1}$ with*

$$\gamma = \min(1 - f, cf \sin(\pi/p)), \quad \mathcal{A} = p(1 + f) + f(q + \frac{p}{2}). \quad (14)$$

Then $\text{thick}(K_d) \geq \rho_d = \gamma^d$ and $L(K_d) \leq 2\pi\mathcal{A}^d$ for all d , hence $\text{Rop}(K_d) \leq 2\pi\Lambda_{\text{Rop}}^d$ with $\Lambda_{\text{Rop}} = \mathcal{A}/\gamma$. For $(2, 3)$ and $f = 1/2$: $\mathcal{A} = 5.00$, and $\Lambda_{\text{Rop}} = 20.0$ with $c = 1/2$ ($\gamma = 0.250$), $\Lambda_{\text{Rop}} = 10.0$ with $c = 1$ ($\gamma = 0.500$).

Proof. Induction on d . If $\text{thick}(K_{d-1}) \geq \rho_{d-1}$ then $r_d = f\rho_{d-1} \leq f \text{thick}(K_{d-1})$, so (H_c) applies and $\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, cr_d \sin(\pi/p)) \geq \min(\rho_{d-1} - f\rho_{d-1}, cf\rho_{d-1} \sin(\pi/p)) = \gamma\rho_{d-1} = \rho_d$. By Lemma 3.3 with $\tau \geq \rho_{d-1}$, $L(K_d) \leq pL(K_{d-1})(1 + f) + f\rho_{d-1}(2\pi q + \pi p)$; since $\rho_{d-1} \leq \text{thick}(K_{d-1}) \leq L(K_{d-1})/2\pi$, this is $\leq L(K_{d-1})[p(1 + f) + f(q + p/2)] = \mathcal{A}L(K_{d-1})$, and $L(K_0) = 2\pi$. For $(2, 3)$, $c = \frac{1}{2}$, $f \leq \frac{1}{2}$, the hypothesis is proved at $d = 1$ (Theorem 3.12(c)) and at $d = 2$, where $r_2 = f\rho_1 \leq f \text{thick}(K_1)$ (Proposition 3.19). $\square$

Remark 3.23 (on the constants). Λ_{Rop} is generous ((H_c) with $c = 1/2$ is twice weaker than the observed $\tau_d \approx r_d$). Numerically the growth factor per depth is 4.06, 4.00 at $f = 1/2$ (Table 2) against $\Lambda_{\text{Rop}} = 20.0$; along the chain of the corollary itself ($r_d = f\rho_{d-1}$, `experiments/corollary_chain.py`) the polygons give $L/\tau = 38.4, 256.5, 2051.9$ against $2\pi\Lambda_{\text{Rop}}^d = 126, 2513, 50265$, and $\tau_d/\rho_d \geq 1.66$ for $f \in \{0.25, 0.35, 0.5\}$. We make no claim that any configuration in the family is stationary or optimal for ropelength, and none that the recursion is a “flow law” for tight recursive knots: it bounds one explicit construction.

Remark 3.24 (lower bounds, quoted). [literature] Buck and Simon [2] prove $\text{Rop}(K) \geq c_{\text{BS}} \text{Cr}(K)^{3/4}$ for a universal $c_{\text{BS}} > 0$, and Cantarella, Kusner and Sullivan [3] prove that ropelength minimisers exist in every knot and link type, are $C^{1,1}$, and satisfy further lower bounds. These need $\text{Cr}(K_d)$. K_2 is a 2-cable of the trefoil (Lemma 3.2), which is adequate; by Kalfagianni and McConkey [8] a cable with q strands of an adequate knot K has crossing number $\geq q^2 \text{Cr}(K) + 1$, so $\text{Cr}(K_2) \geq 13$ (their exact value $4\text{Cr}(K) + 1$ covers the slopes $\pm 5, \pm 7$ of the trefoil, not the slopes 9, 11 of K_2). For $d \geq 3$ only general satellite bounds are available, such as $\text{Cr} \geq 10^{-13} \text{Cr}(\text{companion})$ [9]. For $K_1 = T(2, 3)$ the best known numerical ropelength is about 32.7 (numerical, uncertified; [3, 12]); our depth-one polygon at $f = 1/2$ gives 38.4, i.e. the construction is not close to tight even at depth one.

Conjecture 3.25 (a threshold for $\text{thick}(K_d) = r_d$). [conjectural] For the pattern $(2, 3)$ there are thresholds $f_*(d) > 0$ with $\inf_d f_*(d) > 0$ such that for $f \leq f_*(d)$ and every choice of the phases $\varphi_1, \dots, \varphi_d$ (equivalently, of the frame normalisations) every doubly critical pair of K_d other than the antipodal pairs of one normal disc is at distance $> 2r_d$ and $\min\text{Rad}(K_d) > r_d$; hence $\text{thick}(K_d) = r_d$ (the inequality $\leq$ is Proposition 3.5). At $d = 2$ the part $\min\text{Rad}(K_2) > r_2$ is proved for $f \leq 0.484375$ (Proposition 3.19). Evidence, at the grid values $f \in \{0.25, 0.35, 0.5\}$ only: $f_*(1) \in [0.35, 0.5)$, and $\tau_d = r_d$ at $d = 2, 3$ for $f \leq \frac{1}{2}$, at $f = 0.25, 0.35$ for the default phases and at $f = \frac{1}{2}$ for every phase tested, with polygonal margins $\geq 1.37\%$ ($d = 2$) and $\geq 0.233\%$ ($d = 3$, joint scan), smooth values being slightly smaller (Section 4).

Remark 3.26 (growth rate). If $\text{thick}(K_d) = r_d$ for all large d and $\min\text{Rad}(K_{d-1})$ stays bounded below, then $\text{Rop}_d / \text{Rop}_{d-1} = (L_d/L_{d-1})(\tau_{d-1}/\tau_d) \rightarrow p/f$ by Lemma 3.3. The measured ratios (4.06, 4.00 at $f = 1/2$ against $p/f = 4.00$; the other chains from Table 2) agree: the rate is a consequence of the conjecture, not an independent constant.

4 Experiments

Criteria. The protocol is in Appendix A. Four criteria are checked: (C1) resolution: Rop and τ change by less than 1% between N_0 and $2N_0$; (C2) $L(K_d) \leq$ the right-hand side of (6) evaluated with the measured $L(K_{d-1})$, $\tau(K_{d-1})$ and α_d ; (C3) $\tau(K_d) \geq \rho_d$ of (13) with $c = 1/2$ and the measured $\tau(K_{d-1})$, for $f \leq 1/2$; (C4) τ_{pt} agrees with τ to within 1%. The plan written before any code (`PLAN.md`) fixed the resolution check at $d = 1, 2$ without a threshold, the length bound and the form of (13) with an unspecified constant; the 1% thresholds, (C1) at $d = 3$ and (C4) were fixed when the manuscript was written, and $c = 1/2$ in (C3) after a first run with $c = 1$ had failed at $(f = 1/2, d = 1)$: (C3) is a post hoc check.

Table 2: All chains at the finest resolution ($N_0 = 1024$; 512 for $(3, 2)$). $r_d = f\tau_{d-1}$; $\rho_d^{(1)}$ is (13) with $c = 1$, so $\tau_d/\rho_d^{(1)} < 1$ marks a failure of $c = 1$; slope = $q + pn$ (Lemma 3.2). † : $f = 0.6$ lies outside $r \leq \tau/2$ (probe only).

(p, q)	f	d	N	r_d	L	τ	L/τ	$\tau_d/\rho_d^{(1)}$	Wr	slope
(2, 3)	0.25	1	2048	0.2500	13.444	0.2500	53.8	1.000	3.13	3
(2, 3)	0.25	2	4096	0.0625	26.911	0.0625	430.6	1.000	15.25	9
(2, 3)	0.25	3	8192	0.0156	53.822	0.0156	3444.6	1.000	63.50	33
(2, 3)	0.35	1	2048	0.3500	14.271	0.3500	40.8	1.000	3.26	3
(2, 3)	0.35	2	4096	0.1225	28.608	0.1225	233.5	1.000	15.49	9
(2, 3)	0.35	3	8192	0.0429	57.219	0.0429	1334.6	1.000	63.98	33
(2, 3)	0.5	1	2048	0.5000	15.949	0.4158	38.4	0.832	3.52	3
(2, 3)	0.5	2	4096	0.2079	32.349	0.2079	155.6	1.000	17.99	11
(2, 3)	0.5	3	8192	0.1040	64.730	0.1040	622.7	1.000	74.96	39
(2, 3)	0.6 †	1	2048	0.6000	17.307	0.3401	50.9	0.850	3.71	3
(2, 3)	0.6 †	2	4096	0.2041	34.941	0.1367	255.6	1.005	18.36	11
(2, 3)	0.6 †	3	8192	0.0820	69.893	0.0820	852.1	1.500	75.70	39
(3, 2)	0.5	1	1536	0.5000	20.000	0.4191	47.7	0.968	3.90	2
(3, 2)	0.5	2	4608	0.2096	60.087	0.1813	331.4	0.999	39.66	14
(3, 2)	0.5	3	13824	0.0906	180.270	0.0785	2296.5	1.000	362.97	122
(2, 5)	0.5	1	2048	0.5000	20.420	0.2822	72.4	0.564	6.87	5
(2, 5)	0.5	2	4096	0.1411	41.116	0.1411	291.4	1.000	32.66	19
(2, 5)	0.5	3	8192	0.0705	82.273	0.0705	1166.2	1.000	136.30	71

Results. Table 2 gives all patterns at the finest resolution (plots of the growth and of the curves: `figures/fig_rop.pdf`, `fig_curves.pdf`). All four criteria hold: (C1) the maximal relative change is 0.03% for Rop and 0.03% for τ over all 12 chains (per-level values in `results/tables.md`); (C2) holds in 36 of 36 levels; (C3) holds in all 9 default levels with $f \leq 1/2$, with margin $\tau_d/\rho_d \geq 1.66$ (and ≥ 1.13 including the other patterns; $c = 1$: see after Hypothesis 3.21); (C4) the maximal discrepancy is 0.01%. In every level τ is determined by $\text{dcsd}/2$, never by minRad (36 of 36; $\text{minRad}/r_d \geq 1.15$ throughout).

Three further observations. First, for $f \leq 0.35$ the measured $\tau(K_d)$ equals r_d to four digits at every depth (the antipodal strands in a normal disc are the doubly critical pair, consistent with Proposition 3.5 and Conjecture 3.25), so $\text{Rop}_d = L_d/r_d$ and the recursion $\tau_d = f\tau_{d-1}$ is exact; for $f = 1/2$ at depth one $\tau_1 = 0.4158 < r_1 = 0.5$, realised by the pair across the hole of the torus described in Section 3, while at depths 2, 3 again $\tau_d = r_d$, but narrowly: the next-closest doubly critical vertex pairs (base points at arc $\geq \pi\tau$) are only 1.4% and 2.1% farther than $2r_d$ (1.4% and 2.0% at $N_0 = 512$; 103% and 109% at $f = 0.35$), so $f = 1/2$ is close to the threshold of Conjecture 3.25. Unlike L and τ , these margins depend on the phases (Definition 2.1): over 12 values of $\varphi_2 \in [0, \pi)$ the $d = 2$ margin ranges from 1.37% to 21.3% (period $\pi/3$, from the symmetry of K_1 ; the default is at the minimum), with $\tau_2 = r_2$ and $L(K_2)$ constant to 10^{-5} (`experiments/phase_margin.py`). At $d = 3$ a joint scan of $(\varphi_2, \varphi_3) \in [0, \pi)^2$ (594 pairs at $N_0 = 256$: an 18×24 grid and two local refinements; `experiments/joint_phase_sweep.py`, a frozen copy of the script of `theory/`) gives margins from 0.233%, at $(\varphi_2, \varphi_3) = (1.058, 0.671)$ (0.232% at $N_0 = 512$), to 13.2%, never negative, with $\tau_3 = r_3$ throughout; the minimum over φ_3 has period $\pi/3$ in φ_2 to 0.05 points. These are grid evaluations on vertex pairs, not bounds, and biased upwards (vertex pairs overestimate the distance near a critical pair; the minimum can fall between grid nodes): a smooth construction (spectral Bishop frame, Newton-refined critical pairs of all types; `experiments/smooth_margin.py`, adapted from the round-4 review) gives 1.369% at $d = 2$ (default phases; polygonal 1.4% at $N_0 = 512$) and, at $d = 3$, 2.016% at the default phases and 0.214% at the minimiser of the joint scan. So $\tau_d = r_d$ at $f = \frac{1}{2}$ for every phase tested, at $d = 3$ only narrowly. In the same census (`experiments/classify_pairs.py`, 18 levels of the default family, $f \leq 1/2$, $N_0 \in \{256, 512\}$) no doubly critical vertex pair on the same strand with base arc $< \pi\tau$ and distinct base points occurs; a sign-change test that also detects saddles and maxima of the distance finds same-strand doubly critical pairs only at $d = 1$, with $\delta \geq 1.96 > \delta_{\text{turn}}$ and distance $\geq 3.70r_1$ at $f = 1/2$, and none at $d = 2, 3$. Second, the writhe of the depth-one configuration crosses the half-integer 3.5 inside the grid ($\text{Wr}(K_1) = 3.127, 3.257, 3.518, 3.710$ for $f = 0.25, 0.35, 0.5, 0.6$; the crossing is at $r_1^* = 0.4904$ by bisection), so by Lemma 3.2 the frame linking number is 3 for $f \leq 0.35$ and 4 for $f \geq 0.5$ and the depth-two knot type changes across the grid: K_2 is the $(2, 9)$ -cable of the trefoil for $f = 0.25$ but the $(2, 11)$ -cable for $f = 0.5$. The knot-type decisions are robust to the writhe error (Limitation (5); residual of Lemma 3.2 $\leq 1.1 \cdot 10^{-3}$, smallest margin to a half-integer 0.018; details in

results/convergence.md). Third, $f = 0.5$ gives the best value on the grid at depths one and three; $f = 0.6$ is worse at every depth; f was not optimised.

5 Claims and their status

Claim	Status
K_d embedded if $r_d < \text{thick}(K_{d-1})$; $\text{Lk} = \text{Wr} - \alpha/2\pi$, K_d the $(p, q + pn)$ -cable; length bounds (6) (Lemmas 3.1–3.3).	proved here (Lemma 3.2 from classical results)
$p = 2$: $\text{thick}(K_d) \leq r_d$, $\text{Rop}(K_d) \geq L(K_d)/r_d \geq 2\pi(p(1-f)/f)^d$ (Prop. 3.5).	proved here (internal to the family)
Same-disc pairs $\geq 2r \sin(\pi/p)$; far pairs $\geq 2(\tau - r)$ (Prop. 3.6). Doubly critical pairs, different / same strand: $\geq \min(2(\tau - r), 2c_1 r)$, $\geq \min(2(\tau - r), 2c'_0 r)$; $c_1, c'_0 \geq \frac{1}{2}$ in $(2, 3)$, all d , no (H_3) (Lemma 3.7, Prop. 3.10, Cor. 3.11).	proved here; refined values (0.6117, Table 1) are grid values
$\text{minRad}(K_d) \geq 1/\bar{\kappa}_d$, $\geq 1/\hat{\kappa}_d$ (Prop. 3.8).	proved under (H_3) ; unconditional at $d = 1$
$(2, 3)$, $f \leq \frac{1}{2}$: $\frac{1}{2} \text{dcsd}(K_d) \geq r_d/2$ for all d ; $\text{thick}(K_1) \geq r_1/2$; for $d \geq 2$, $(H_c) \Leftrightarrow \text{minRad}(K_d) \geq r_d/2$ (Thm. 3.12).	proved here
$\nu, \kappa_{\max}$ of K_d from the data of K_{d-1} ; for $d \geq 3$, (H_c) under the growth condition (12); no locally bounded function of the (H_3) data of an arbitrary base bounds $\mu(K_d)$ (Lemma 3.14–Prop. 3.17).	proved here; checked by an independent internal referee (round 4)
$\text{minRad}(K_2) \geq 0.842 r_2$, so $\text{thick}(K_2) \geq r_2/2$, for every $f \leq \frac{1}{2}$ and all phases; $\text{minRad}(K_2) > r_2$ for $f \leq 0.484375$ (Prop. 3.19).	computer-assisted proof (interval arithmetic); not yet checked by an independent referee
(12) at $d = 3, 4$; (10), (11) on the polygons; $\text{thick}(K_3) \geq r_3/2$ (Remark 3.18, Table 1).	grid evaluation, polygon inputs
No $c > 0$ depending only on (p, q, f) works for every base (Remark 3.13).	proved here
(H_c) , $(2, 3)$, $c = \frac{1}{2}$, $d \geq 3$; (H_3) uniformly in d .	open; numerically true in 9 levels, margin ≥ 1.66
$\text{Rop}(K_d) \leq 2\pi\Lambda_{\text{Rop}}^d$, $\Lambda_{\text{Rop}} = \mathcal{A}/\gamma$ (Cor. 3.22).	proved; unconditional for $d \leq 2$ $((2, 3), c = \frac{1}{2})$, conditional on (H_c) for $d \geq 3$
Polygons with $L/\tau = 38.4, 155.6, 622.7$ ($f = 0.5$, $d = 1, 2, 3$), stable to 0.03%.	computed (polygonal proxy, uncertified)
K_2 changes knot type between $f = 0.35$ and 0.5 ($r_1^* = 0.4904$).	computed; consequence of Lemma 3.2
$\text{Rop} \geq c_{\text{BS}} \text{Cr}^{3/4}$; tight knots; $\text{Cr}(K_2) \geq 13$; satellites.	literature [2, 3, 8, 9]
$\text{thick}(K_d) = r_d$ for $f \leq f_*(d)$ and all phases; then $\text{Rop}_d / \text{Rop}_{d-1} \rightarrow p/f$ (Conj. 3.25).	conjectural; at $f = \frac{1}{2}$ polygonal margins $\geq 1.37\%$, $\geq 0.233\%$ over the phases tested (smooth: 0.214% at $d = 3$)
Stationarity, optimality, a recursive flow law.	not claimed

6 Limitations and next steps

Limitations. (1) The thickness lower bound ($p = 2$, default family) is proved for all doubly critical pairs at every depth and completely at $d = 1$; for $d \geq 2$ the curvature of K_d needs (H_3) , whose constants are only measured on the polygons; their propagation in d is partial (Corollary 3.16 for $d \geq 3$ under a growth condition; Proposition 3.17); $d = 2$ is settled by a computer-assisted proof (Proposition 3.19), not sharp and not yet checked by an independent referee; $d \geq 3$ is open. (2) The polygonal τ of (4) (vertex pairs, local-minimum test) is supported by τ_{pt} , the segment check and stability in N , but is not certified. (3) Nothing is optimised; the family (pattern angle linear in t , $\varphi_d = 0$, the frame normalisation of Definition 2.1) is one of many, and the margins of Conjecture 3.25 depend on the phases, which were scanned only at $f = \frac{1}{2}$ (at $d = 3$ on a grid with local refinement); for $p \geq 3$ the thickness lower bound is treated only in Proposition 3.6. (4) Only the thresholds $c_1 \geq \frac{1}{2}$, $c'_0 \geq \frac{1}{2}$, $1/(r\bar{\kappa}_1) > 0.561$, the closed forms of Corollary 3.11 and the constants of Corollary 3.16 are analytic, and the bound of Proposition 3.19 is computer-assisted; every other constant of Section 3 is a grid infimum or supremum (c_1, c'_0 with a Lipschitz constant estimated on the grid itself, $\hat{\kappa}_d$ with a neighbour-increment correction, the v0.2 constant c_0 without correction), with polygon-measured inputs for $d \geq 2$; only Proposition 3.19 uses interval arithmetic. (5) Writhe and holonomy are midpoint-rule sums, stable to $1 \cdot 10^{-4}$ at depth one but only to 0.031 at depth three of $(3, 2)$; the frame linking number for $f = 0.5$ is decided by a margin of 0.018; an exact polygonal writhe is not implemented. (6) The only lower bound specific to the family is Proposition 3.5; knot-type bounds are limited by what is known of $\text{Cr}(K_d)$ (Remark 3.24).

Next steps. (1) Make (H_3) uniform in d : by Proposition 3.17 this needs control of all derivatives, e.g. an induction on analyticity in a complex strip around the real axis (the Bishop frame solves a linear ODE with analytic coefficients); with Corollary 3.16 this would prove (H_c) with $c = 1/2$ for $(2, 3)$ at every depth $d \geq 3$, and Corollary 3.22 unconditionally for every depth. At $d = 3$, Corollary 3.16 needs bounds

on $\nu(K_2)$, $\mu(K_2)$, which depend only on the local data of K_1 (up to fourth derivatives), on r_2 , m_2 and on the angle of U in the normal plane: an interval computation over boxes in (s, angle, r_2) , with the angle over the whole circle (every phase), would settle $d = 3$ without integrating the frame. (2) Certify: interval arithmetic for c_1 , c'_0 , $\hat{\kappa}_d$ ($d \geq 3$) and for minRad , dcsd of a piecewise-circular interpolant; exact polygonal writhe (Levitt; Klenin–Langowski). (3) A constrained length minimisation at fixed $\tau \geq 1$ with a certified knot-type check, for $d = 1, 2$ (a curve-shortening probe of `v0.1`, `experiments/shrink.py`, gave nothing usable). (4) Optimise f_d per depth and the pattern angle in arclength; treat $p \geq 3$ (the bound δ_{R2} is general, the a priori bounds use $m \leq 2$ and $m_1 = 3/2$). (5) Match the conventions of [8] with Lemma 3.2 to obtain $\text{Cr}(K_2)$ and attach a Buck–Simon bound to each depth-two configuration.

A Reproducibility and protocol

Protocol. `experiments/recursive_knots.py` builds the polygon K_0 with N_0 vertices, computes the closed Bishop frame by discrete parallel transport (Rodrigues rotation taking T_i to T_{i+1} ; initial normal as in Definition 2.1, with the central-difference tangent; holonomy = signed angle between transported and initial normal, removed by the twist $-\alpha i/N$) and produces K_d with $p_d N_{d-1}$ vertices by (3) at the vertices (t is the vertex index). At each depth it measures L , minRad , dcsd , τ , τ_{pt} , Rop , writhe, holonomy and $\text{Lk} = \text{rint}(\text{Wr } K_{d-1} - \alpha_d/2\pi)$ with its residual, and sets $r_{d+1} = f \tau(K_d)$. Grid: $(2, 3)$ with $f \in \{0.25, 0.35, 0.5, 0.6\}$, $(3, 2)$, $(2, 5)$ with $f = 0.5$; depths 1–3; $N_0 \in \{512, 1024\}$ ($\{256, 512\}$ for $(3, 2)$). Deterministic; 184s wall-clock, Python 3.11.15, NumPy 2.4.6. The auxiliary scripts (critical pairs and segment check; constants of Section 3 and every intermediate inequality, with frozen copies of the scripts of `theory/` and SHA-256 in `results/FROZEN_THEORY_SHA256.txt`; census of doubly critical pairs; chain of Corollary 3.22; scans of the phases; smooth margins and (H_3) data; the computer-assisted proof of Proposition 3.19, frozen with SHA-256, and the author’s check of it) and their run times are listed in the README. `make_numbers.py` writes `numbers.tex` and the table body from `results/*.json` only. Every number in this text is a macro from `numbers.tex`.

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